GREAT LAKES/ PROJECT REPORT

Wetland Vegetation Monitoring in Cootes Paradise

Measuring the Response to a Fishway/Carp Barrier

by Tyler Smith, Jeremy Lundholm and Len Simser

Monitoring a wetland restoration at Cootes Paradise yields valuable information about which monitoring methods work and whether planting is really necessary.

Ecological restoration is an iterative process. Efforts to address one problem often result in the discovery of new problems. The restoration technique used may be found to be inadequate for the task at hand. Alternatively, successful techniques may uncover secondary problems that had previously been masked by the primary environmental stresses. Evaluation of these scenarios is dependent upon a sound monitoring program.

Vegetation monitoring techniques can include those used by plant ecologists, but often the specific needs of restorationists are not met by the general set of vegetation sampling protocols. The scale and unique circumstances of a given restoration project need to be taken into consideration when designing monitoring programs. What follows is an account of efforts over the past seven years to monitor changing wetland vegetation in a large coastal marsh in response to new restoration techniques.

Cootes Paradise is a 625-acre (250-hectare) wetland owned and managed by Royal Botanical Gardens (RBG). Royal Botanical Gardens is a non-profit organization dedicated to botanical research and education, with special emphasis on the social and ecological value of plants. The Cootes Paradise sanctuary is located at the west end of Lake Ontario, bounded by Hamilton Harbour on the east and the town of Dundas on the west (Figure 1). The wetland area is a drowned re-entrant valley associated with the Niagara Escarpment. Most of the area of the wetland is covered with open water or seasonal mudflats, although historic accounts indicate that the extent of emergent vegetation has at times approached 100 percent coverage (Painter and others, 1989). The wetland has become seriously degraded during the past 200 years. Factors implicated in its decline include: sewage deposition from residential and industrial sources; destructive feeding habits of common carp (*Cyprinus carpio*), an invasive exotic fish; marsh alteration related to the Desjardin Canal; and water level manipulation on the Great Lakes in service of shipping concerns. The implications for the ecosystem have been a decrease in the abundance and diversity of the vegetation community, a related decline in fish and wildlife populations, and a decline in water quality (Chow-Fraser and others, 1998).

Despite the ecological decline of Cootes Paradise, it remains one of the most important natural areas in the Great Lakes Region, providing critical migratory and nursery habitat for birds, fish and amphibians (Painter and others, 1989). In recognition of these functions, the authors of the Fish and Wildlife Habitat Restoration Project for the Hamilton Harbour Remedial Action Plan identified it as a

Ecological Restoration, Vol. 19, No. 3, 2001

keystone area (Trotter and others, 1998). The remedial action plan was prepared in 1992, and restoration efforts are ongoing. They have included the construction of combined sewage overflow tanks, watershed stewardship programs in cooperation with private landowners, wetland vegetation plantings, lobbying the International Joint Commission to restore natural water levels, and the installation and operation of the fishway/carp barrier to exclude carp from the marsh.

The goals of the Cootes Paradise restoration include increasing the diversity and abundance of wetland vegetation. Consequently, vegetation monitoring has been critical in assessing the results of restoration initiatives to date. Due to the scale and complexity of the situation, we have used a number of different techniques to measure the plant communities of Cootes Paradise. The data we have collected in surveys during the past seven years show the restoration to be an overwhelmingly positive venture. In addition, experience gained in the process has allowed us to evaluate several methods for the long-term monitoring of emergent and submergent plant communities.

Baseline Data

The vegetation monitoring program has benefited from the perspective provided by a number of historical surveys. The first botanical surveys of Cootes Paradise were conducted in the 1940s, and the resulting checklists (Judd, 1950) and maps (Wragg, 1949) have become the yardstick by which progress in the restoration is measured. Ecological degradation of the marsh was first quantified by botanical surveys in the early 1970s (Simser, 1982). Prior to the major restoration initiatives of the present project, another set of maps and checklists was compiled in 1993 (Pomfret and Benkhuysen, 1993).

In addition to directed botanical surveys, air photos of Cootes Paradise have been digitized for use in GIS analysis. These photos, dating back to 1934, were not taken with the express intention of monitoring wetland communities. They have, however, provided invaluable insight into vegetation dynamics over the past 66 years (Chow-Fraser and others, 1998).

Restorative Measures: Land Stewardship, Carp Removal, and Wetland Planting Program

Cootes Paradise has been degraded by land use practices throughout the watershed as well as by point-source contamination. Today, the Hamilton-Halton Watershed Stewardship Project addresses issues related to marsh water quality and upstream erosion. This program, coordinated cooperatively by the Hamilton Region Conservation Authority and Conservation Halton, encourages private landowners to adopt ecological land management practices, particularly where streams cross their properties. Sewage discharge into the marsh is being partially redirected into combined sewer overflow holding tanks. Ongoing research is directed at addressing the impact of sewage discharge from the Dundas Sewage Treatment Plant. While we expect to see some immediate benefits to water quality as a result of these initiatives, we hope that the true ecological gains will occur in the slow recovery of the marsh over coming decades.

In contrast, the impact of carp on the wetlands is direct and immediate. The fishway/carp barrier (Figure 2) was constructed in 1996 and was operational for the 1997 season. The fishway grates allow for free water exchange between the marsh and the harbor and the unimpeded passage of fish less than 2 inches (5 cm) wide in and out of the marsh. Larger fish, including sexually mature carp, swim into large metal cages ("baskets") during the spring spawning migration. The baskets are lifted and emptied three to ten times weekly, following seasonal fish migration cycles. When the baskets are emptied, native fish species are released into the marsh, while carp are returned to Hamilton Harbour. After spawning most fish migrate back out to the harbor, which for larger individuals again requires their being lifted through the fishway. Culling the carp is not possible because they contain PCB-contamination levels in excess of provincial guidelines for landfill or compost (Hahn, 1997).

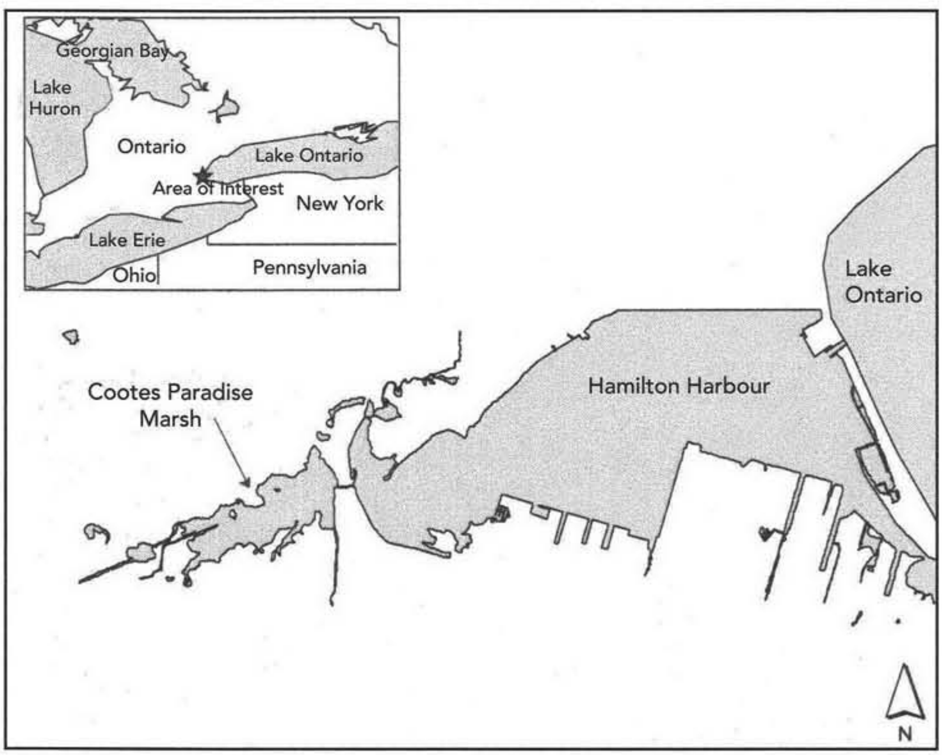

Figure 1. Location of Cootes Paradise relative to Lake Ontario

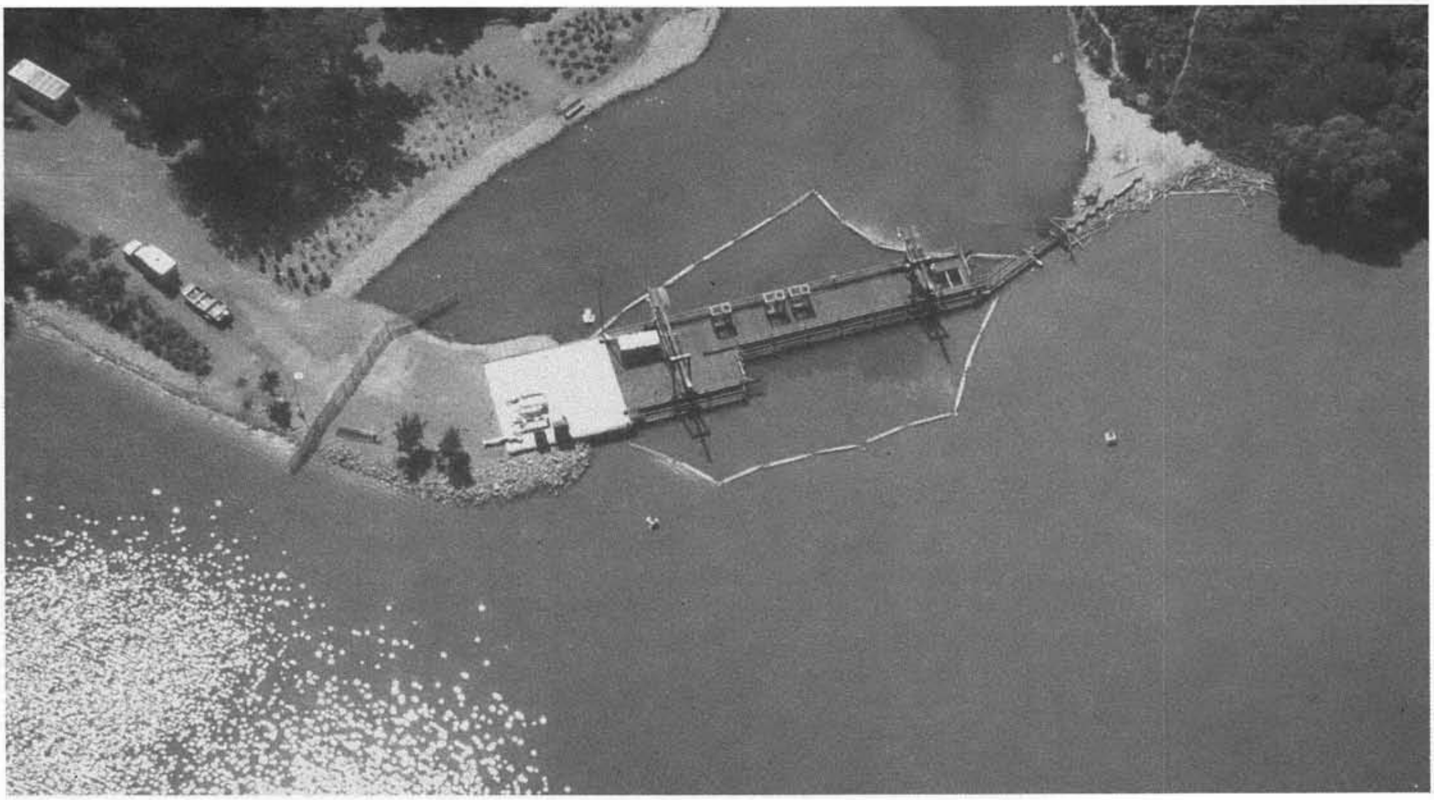

Figure 2. Cootes Paradise Fishway/Carp Barrier as seen from the air. *Photos courtesy of Tyler Smith*

We expected to see measurable water quality and habitat improvements in the marsh within the first few years following carp exclusion. With three years of post-exclusion data we have found that there has indeed been a significant positive response, including increases in plant density and diversity, improved water clarity, and reductions in suspended sediment and dissolved phosphorus.

Restoration of Cootes Paradise has not been limited to the reduction of negative influences on the indigenous vegetation. We have also tried to accelerate the natural recovery of plant communities with a planting program. Initially, we made use of plant material from Wisconsin due to the paucity of commercially available indigenous stock at that time. As the program matured we switched to locally-collected seed and developed our own plant nursery to supply our plant material. Both volunteer and staff plantings were carried out, emphasizing areas of the marsh where water depth and exposure were conducive to emergent plants. In addition, we cordoned off planted areas in a partially successful attempt to limit Canada geese (*Branta canadensis*) from uprooting the young plants. The benefits of the planting included increased community awareness (Chow-Fraser and Lukasik, 1995) and a better appreciation of aquatic plant ecology, both by staff and volunteers. The loss of planted material to resident populations of Canada geese suggested that these animals may be inhibiting the recovery of the vegetation in general.

Emergent Plant Monitoring Methods: Aerial Photos, GPS, Transects and Photo-Monitoring

We identified two aspects of the emergent plant community as critical elements of the restoration: diversity and total area covered by plants. In 1998, we compiled a species list after intensive surveys were conducted in emergent plant communities. This list is compared to previous surveys in Table 1. We have used several techniques to quantify the spatial extent of emergent vegetation, including transect surveys, ground level photomonitoring, aerial photography, and GPS mapping. We have also compiled historic data using air photos. A search of local records yielded 14 photos taken since 1934 (Lavender, 1987). The extent of the emergent vegetation visible in these photos was digitized and is now stored in a GIS database. This information has proven invaluable in describing long-term trends in vegetation cover.

Aerial photo-monitoring requires careful planning. Ideally, photos should be taken late in the growing season, at the time when the plant communities have reached their greatest extent. Air photos in the spring, on the other hand, may obscure vegetation due to high water levels. Another consideration is the use of orthorectified photos. Orthorectified photos are referenced to surveyed points on the ground, reducing the degree of distortion inherent in the photographic process. This greatly improves the accuracy of measurements taken from the photos, but with a tenfold increase in cost. We have opted to use orthorectified photos at Cootes Paradise

Table 1. Emergent Plant Species in Cootes Paradise. Species in ALL CAPS are exotic.

Species	1950	1993	Post Fishway
Equisetum fluviatile	+		
Alisma plantago-aquatica	+	+	+
Sagittaria latifolia	+	+	+
Acorus americana	+		+
Calla palustris	+	+	+
Carex lacustris		+	+
Carex retrorsa			+
Scirpus acutus		+	
Scirpus fluviatilis		+	+
Scirpus pungens	+		
Scirpus validus	+	+	+
Zizania aquatica	+		
Sparganium emersum	+		
Sparganium eurycarpum	+	+	+
TYPHA ANGUSTIFOLIA		+	+
Typha latifolia	+	+	+
TYPHA X GLAUCA		+	+
IRIS PSEUDACORUS		+	+
Iris versicolor	+	+	+
Polygonum amphibium	+	+	+
Rumex orbiculatus			+
Rumex verticillatus		+	+
Rotentilla palustris	+		+
Decodon verticillatus	+	+	+
LYTHRUM SALICARIA	+	+	+
Cicuta bulbifera	+		+
Cicuta maculata			+
Sium suave			+
Total Native/Introduced	16/1	13/4	19/4

because the greater accuracy allows us to take advantage of our developing GIS database. We have partnered with other agencies in the area to share costs and resources, although budget constraints have limited the frequency of air photo monitoring.

As a more flexible alternative to aerial photography, we have made extensive use of a GPS receiver to produce accurate maps of vegetation stands, as well as other components of the restoration. Accurate to within 3 feet (1 m) in most situations, the GPS data is comparable to our digitized air photos. There are, however, three factors to consider when contemplating the use of GPS: 1) signal reception, 2) accessibility, and 3) cost. First, GPS units are generally unreliable under tree canopies. This is not a problem in open marshland, but forests or even steep ravines can prevent signal reception. Second, to collect GPS data you must be able to get to the area you wish to measure. Some areas of Cootes Paradise can only be mapped in winter, when the deep, loose organic sediment has frozen. Third, as with aerial photography, GPS can be expensive since a unit with a 3-foot accuracy will cost \$15,000 (CDN) or more. As an alternative, many firms rent GPS receivers. We completed a basic map of Cootes Paradise, with 126 acres (51 hectares) of vegetation, in one week. Of course, more detailed maps require more time in the field.

In addition to aerial photography, we established permanent transects throughout the marsh. These transects were used to record the distance across the emergent stands, giving an estimate of the expansion or contraction of the vegetation along a line perpendicular to the shoreline. We have abandoned this monitoring method, however, because it became increasingly difficult to accurately measure the distance across long stands of vegetation. For example, we found that in traversing a 245-foot (75-m) cattail (*Typha* spp.) stand distance measurements could be off by as much as 10 feet (3 m). This is because of the difficulty in maintaining a straight line through a dense stand of 10-foot (3-m) tall plants, even with a crew of three people. By comparison, GPS measurements are much more accurate. The second difficulty with transects became apparent with the large increase in emergent growth resulting from recent low water levels in Lake Ontario. We established all transects in a north-south alignment without considering the depth gradients of the marsh. This was an unfortunate oversight because in locations where the transects run parallel to the depth contours they have in effect become a presence/absence metric of limited value.

The last technique we use to monitor the emergent vegetation is ground level photo-monitoring. While this is not a quantitative method, it provides an informative visual image that complements the "hard" data collected with the GPS receiver. By establishing permanent photo-stations and taking pictures from the same perspective each year, the resulting slides need little interpretation as qualitative measurements (see Van Horn and Van Horn, 1996 for quantitative photomonitoring techniques). One unforeseen difficulty we encountered was having many of the photo-stations obscured by the huge increase in cattail stands. Consequently, the best locations for photo-stations are those where structural features (boardwalks, pilings) allow us to climb above the surrounding vegetation. The photos from one of these sites are shown in Figure 3.

Results: Emergent Plant Monitoring

Examination of the emergent species inventory results revealed several challenges. First, we found that more species are listed in 1998 than in 1993 or 1950. This appears to indicate that the restoration has had a significant effect on the diversity of the emergent plant community. On reflection, however, the data sets are not easily compared. Neither the area surveyed nor the species considered to be emergents were defined in either of the previous studies. The 1998 survey examined all areas of the marsh. Some species, such as water parsnip (*Sium suave*), were found only in a few locations well removed from the main body of the marsh. It is likely that the previous surveys were more limited in area and may have overlooked these species. On the other hand, some of the species on the list are difficult to categorize. River bulrush (*Scirpus fluviatilis*), for example, is common along the waterline in Cootes Paradise, but in dry years may be considered part of the terrestrial flora. It is difficult to rule out the possibility that previous surveys overlooked species recorded in 1998. Since the 1998 survey was apparently more intensive than previous efforts, it is unlikely that the reverse is true. Any species not recorded in 1998 and present in 1950 have almost certainly been extirpated from the marsh. Four species fall into this category. There is a chance that these species may recolonize the marsh from elsewhere in the watershed as conditions improve. If they do not we will add them to our nursery propagation program for re-introduction.

We can avoid these difficulties in future by documenting our surveys with detailed maps, as we have done for the 1998 checklist. In addition, surveys should include the rationale used to determine which species were included and which were excluded during data collection. Another important tool in diversity monitoring is the preparation of herbarium vouchers. A thorough herbarium collection allows future researchers to examine the raw data of the inventory. This is particularly useful where current research interests differ from those of the original study.

The GPS data, when compared to previous maps, shows a significant increase in the area of emergent vegetation in Cootes Paradise since the installation of the fishway. In 1993, we mapped the extent of emergent vegetation by hand (Figure 4), and it covered 82 acres (33 ha) or 13.7 percent of the entire marsh. The GPS map we made in 1999 (Figure 5) shows 126 acres (51 ha) of emergent plants, or 18.8 percent of the marsh.

We attribute the increase in the area of emergent vegetation to three things: water levels, carp exclusion, and planting programs. As in any wetland the most important factor is water level. In 1993 the average water level in Lake Ontario was 246.2 feet above mean sea level (FSL) (75 meters above mean sea level [MSL]), but in 1999, following several years of drought, the water level averaged only 244.7 FSL (74.6 meters MSL). The low water levels in 1998 and 1999 allowed emergent vegetation to increase dramatically from natural sources. The relationship between periodic low water levels and the expansion of emergent vegetation is well known (see, for example, van der Valk, 1981 and Keddy and Reznicek, 1986).

Despite the acknowledged role of water levels on plant growth, we also observed areas outside of Cootes Paradise where vegetative growth was dramatically different because carp were excluded from the system. We noticed that areas protected from carp supported lush growth of

Figure 3. Photo-monitoring from the same point in Cootes Paradise. a) September 1998, shows the full extent of vegetation growth for the season; b) June 1999, shows cattail (*Typha* spp.) seedlings germinating on the mudflat exposed by low water levels; c) October 1999, shows the full extent of vegetation growth for the season.

emergent vegetation while unprotected areas had no new growth of emergent plants (Figure 6). This suggests that while water level is the primary factor controlling growth of emergent plants, carp can seriously limit plant growth even when water levels are favorable. We credit the significant reduction of carp in Cootes Paradise with enhancing the response of emergent vegetation to favorable water levels.

The third factor influencing emergent plant growth at Cootes Paradise is our planting program. However, when we compare the 1999 vegetation map with the list of species planted in different areas, there is little evidence to support the claim that the new growth is the result of our planting efforts. This is because the overwhelming factor controlling the growth of emergent vegetation is lower water levels. The plantings were on the whole unsuccessful because they coincided with above-average water levels. In an attempt to address this issue, we attempted to artificially control the water levels in 1994 and 1995 by using a portable dam, the "Aquadam." This structure can be floated into place and filled with water. Once installed it forms a water-proof barrier that allows for water to be pumped out of large areas of marsh. We had hoped to plant the temporarily drained sections of the marsh and let the emergents become well-established before allowing the water level to gradually rise again. However, the Aquadam was damaged by vandals before our plans could be completed. This technique may warrant further investigations. Aquadam design specifications and retail distribution information is available at www.aquadam.com/.

In light of the tremendous natural increase in emergent vegetation in response to low water levels in 1998 and 1999, it is unlikely that any amount of planting will significantly increase the marsh vegetation on the whole without some form of water level control. This being the case, we have redirected our plantings of emergents towards areas that are isolated and exposed, and unlikely to become naturally revegetated. By combining planting with the construction of wave-breaks we hope to accelerate the natural revegetation processes.

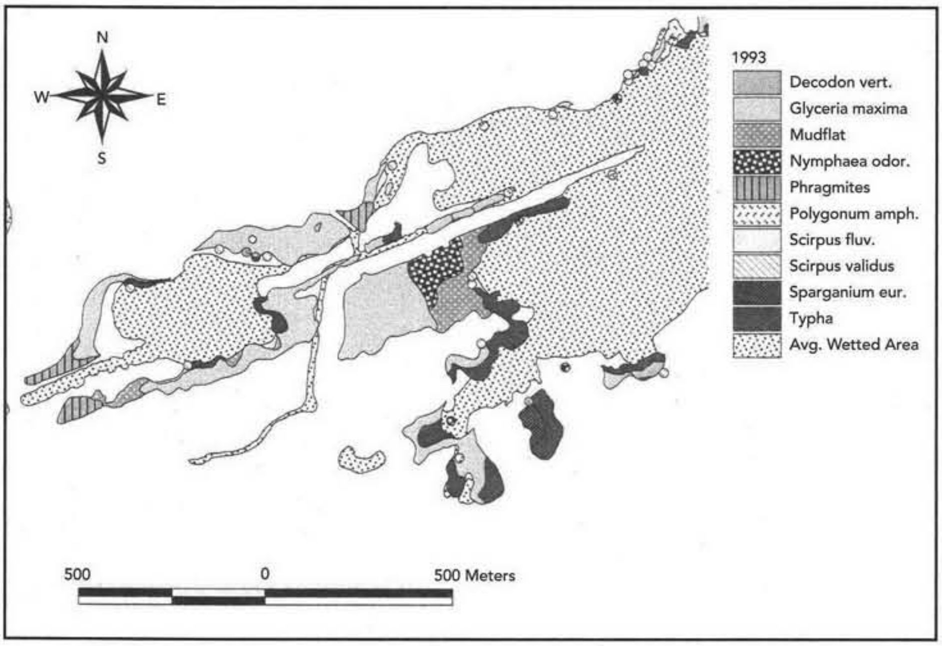

Figure 4. 1993 Cootes Paradise-West End emergent vegetation map shows the extent of emergent vegetation was relatively small.

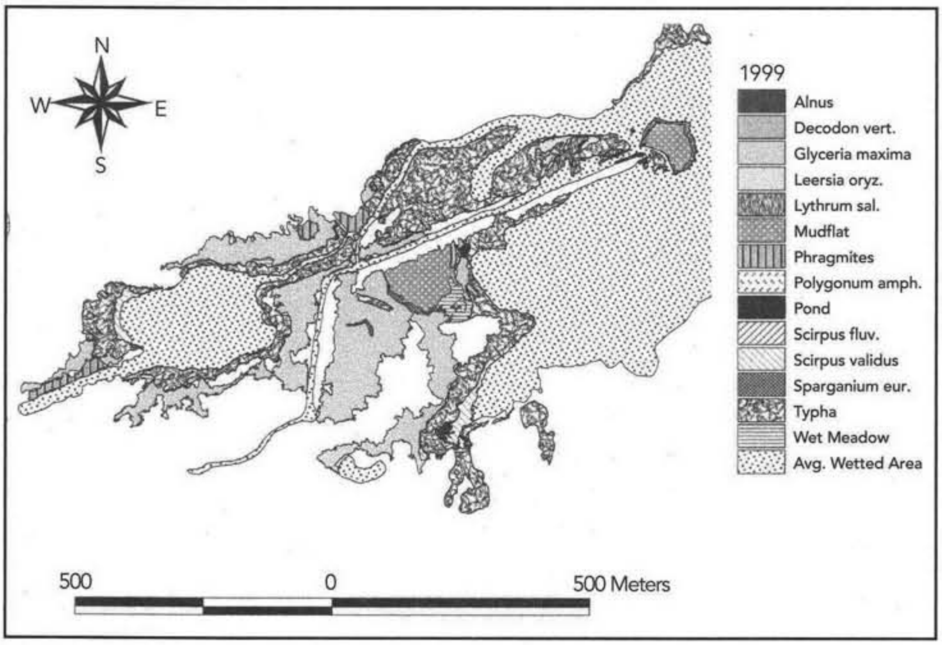

Figure 5. 1999 Cootes Paradise-West End emergent vegetation map shows a significant increase in emergent vegetation as a result of low water-levels and fewer carp.

Submergent Plant Monitoring Methods: Transects and Surveys

The aerial extent of submergent plants is difficult to measure in the field. We decided that it would be more appropriate to monitor the density of submergent plants for different parts of Cootes Paradise. To this end, in 1996 we established eight permanent transects in the marsh (Figure 7). In 1996, 1997 and 1998, we identified and recorded all submergent plants visible inside a 9-foot (3-meter) width centered on the transect. In 1999 we reduced the width of the transect to 20 inches (50 cm) in response to the dramatic and unexpected increase in sub-

Figure 6. Carp exclosure in Grindstone Creek. The dense plant growth behind the barrier (left) is mostly spontaneous growth attributed to the elimination of carp in the area. Similar growth was noted on the barren mudflat (right) early in the year but none of it survived the season.

mergent plants recorded in 1998 (Figure 8). In the most heavily vegetated areas of the marsh in 1998, we counted 3,866 plants in a 570-foot (190-meter) transect. Changing the transect width reduces the possibility of counting error, while still allowing for the calculation of plant densities.

Initially, we measured the transects in June. While the density of submergent plants peaks later in the season, boat access becomes a problem when the water level drops. However, with the tremendous increase in plant densities in 1998 we realized that not only do the number of submergent plants peak in August, but the species in the marsh change through the season. Sago pondweed (*Potamogeton pectinatus*) was the dominant plant in June, while Eurasian milfoil (*Myriophyllum spicatum*) was the most abundant plant in August. Flat-stemmed pondweed (*Potamogeton zosteriformis*) and perfoliate pondweed (*Potamogeton perfoliatus*)—two species of local interest—don't appear in the marsh in any number until July. We decided to address these issues by surveying twice each season. All the transects are monitored in June, and beginning in 1999, any transects still accessible by boat in August are remeasured at that time.

The transect data is presented in Figure 8. The transects are of varying lengths, so total plant densities for the marsh are calculated as the mean of all transects, weighted by the size of the areas corresponding to each transect. The results of these calculations are presented in Figure 9. This is an admittedly coarse estimate. In the future we will be investigating the use of our GPS receiver to collect data from a larger number of smaller transects. We hope this will allow us to produce more accurate density estimates through the use of sophisticated software packages ("raster interpolation") now available in our GIS lab. Increasing the number of transects will also allow us to improve our sampling methods to suit the patchy distribution of the submergent plants.

We compiled a comprehensive species list of submergent plants found in Cootes Paradise from survey work conducted over the course of the 1998 and 1999 seasons. This list is tabulated with the results of previous surveys in Table 2.

Results: Submergent Plant Monitoring

Comparing current and historic submergent plant checklists is far less complicated than is the case with the emergent plant lists. The extent of the emergent plant community may be somewhat subjective, both in terms of species considered and geographic boundaries. Submergent plant communities can be more objectively circumscribed. We define submergent plants as those species that are

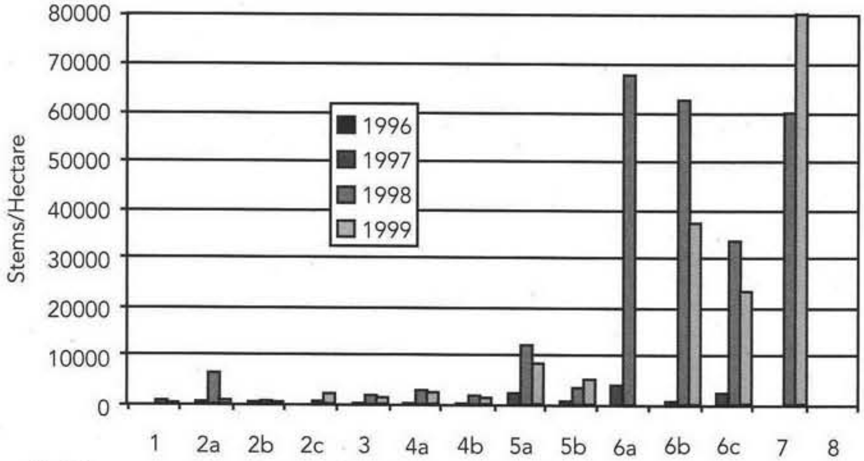

Figure 7. Submergent plant densities by transect. The elevated density for transect 7 in 1999 represents a dense patch of sago pondweed (*Potamogeton pectinatus*) in a very small area. Transect 7 corresponds to less than 1 percent of the total area of the marsh, so this elevated density has little influence on the total submergent densities.

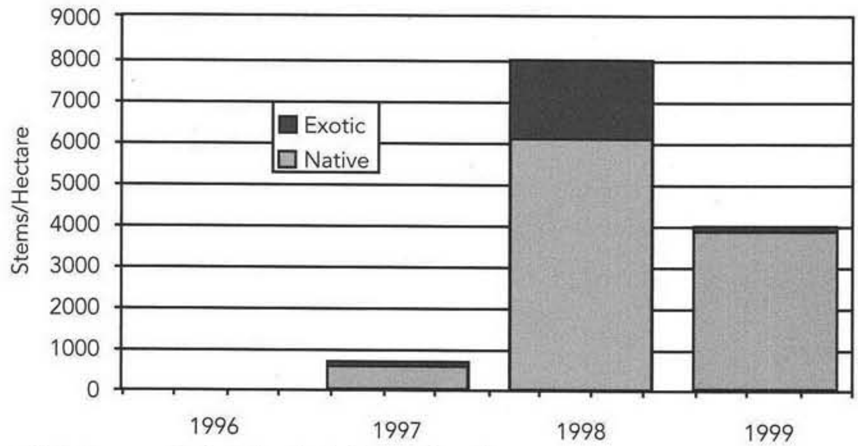

Figure 8. Submergent plant density in Cootes Paradise.

supported throughout most or all of the season by the water-column. This definition narrows the discussion to the 28 species listed in Table 2, including species often referred to as floating-leaved and free-floating. It also limits the geographic extent of the survey to those areas of Cootes Paradise where standing water is present throughout the growing season.

The restricted nature of the submergent plant community in both taxonomic and geographic terms allows for direct comparison between current and historic data sets. As shown in Table 2, eight native species previously extirpated from Cootes Paradise have returned following the installation of the fishway/carp barrier. We have also found three new exotic species. Eurasion water-milfoil has become an invasive concern both in Hamilton Harbour and in the marsh area inside the fishway. Slender naiad (*Najas minor*) and yellow floating-heart (*Nymphoides peltata*) are both new records for Hamilton-Wentworth region. Slender naiad is a halophyte and is not expected to become a serious problem in a freshwater marsh. Yellow floating-heart is known from only a few collections in Ontario. Because it may become an invasive problem, we have removed it by hand, and will continue to watch for it in the future.

The transect surveys show a tremendous increase in submergent densities (Figure 9). We attribute the new abundance of submergent plants in large part to the fishway/carp barrier. We were not expecting to see such a dramatic improvement so soon. Previous seed bank studies indicated that the submergent seed bank was completely depleted and submergent plants would not return unless they were introduced artificially (Westcott and others, 1997). If the seed bank was indeed depleted, the new growth of submergents may have come from dormant rootstock or from propagules dispersing into the marsh from elsewhere in the watershed (Lundholm and Simser, 1999). Our planting program was restricted to emergent and wet meadow plants. It should be noted that while submergent plant densities declined by half between 1998 and 1999, this difference is the result of a 30-year low in water levels that seriously limited the extent of submergent plant habitat and exacerbated the remaining stressors (primarily water quality and nutrient loading). We expect submergent densities will increase again when water levels rise.

Monitoring and Restoration: Some Thoughts

It is now apparent that Cootes Paradise was in considerably better shape than previously assumed. Whillans (1996) attributed the degraded condition of Cootes Paradise to the "failure of recovery mechanisms." Our work shows quite clearly that the natural recovery mechanisms of the marsh are indeed intact. Prior to carp exclusion, some researchers had asserted that the Cootes Paradise basin was deepening as a result of historic stream channelization and that the deeper water the principle factor inhibiting the growth of emergent plants (Chow-Fraser and others, 1998). Bathymetry mapping conducted by RBG staff in 1999 disproved this hypothesis, as the Cootes Paradise basin was shown to have become substantially shallower in the past 50 years. It has also been suggested that wind resuspension and substrate qualities may be more significant than carp in impeding aquatic vegetation (Lougheed and others, 1998). While these and other factors are undoubtedly still impeding the natural functioning of the Cootes Paradise wetland, we can now state with certainty that excluding carp from the system has resulted in substantial improvements to the ecological value of the marsh. These improvements are manifested by increases in abundance and diversity of aquatic and emergent plants.

Future work in Cootes Paradise will be directed at identifying and addressing remaining stresses. We are currently investigating sediment contamination issues in two tributaries and we hope to re-establish natural water cycles in areas where meandering creeks have been channelized within the marsh. The effect of urbanization of the watershed on marsh hydrology is also major concern. As with all urban natural areas in Ontario, we are concerned with the influence of invasive plants and Canada geese on the restoration of the marsh.

Table 2. Submergent Plant Species in Cootes Paradise. This list includes floating-leaved and free-floating plants. Species in ALL CAPS are exotic.

Species	1950	1972	1993	1999
Elodea candensis	+	+	+	+
Vallisneria americana			+	+
Potamogeton berchtoldii	+			
POTAMOGETON CRISPUS	+	+	+	+
Potamogeton foliosus	+			+
Potamogeton natans	+			+
Potamogeton nodosus	+	+		+
Potamogeton pectinatus	+	+	+	+
Potamogeton perfoliatus	+			+
Potamogeton zosteriformis	+			+
Najas flexilis	+			
NAJAS MINOR				+
Zannichellia palustris				+
Lemna minor	+	+	+	+
Lemna trisulca	+			+
Spirodela polyrhiza	+	+	+	+
Wolffia borealis				+
Wolffia columbiana	+	+		+
Heteranthera dubia				+
Nuphar advena	+		+	
Nuphar variegata	+	+	+	+
Nymphaea odorata	+	+	+	+
Ceratophyllum demersum	+	+	+	+
MYRIOPHYLLUM SPICATUM				+
Myriophyllum verticillatum	+		+	
NYMPHOIDES PELTATA	+			
Utricularia vulgaris	+	+		+
Megalodonta beckii	+			
Total Native/Introduced	21/1	10/1	10/1	18/4

The key to any restoration is effective monitoring. A sound monitoring program is the only way to document the successes and learn from the failures of a project. While each project is unique, we believe there are some basic guidelines that can be applied universally in ecological restoration:

1. Collect as much baseline and historical data as possible. We are lucky in that considerable research was conducted in Cootes Paradise in the 1940s, although this aspect of a project is often one that we have the least control over. An accurate depiction of the current and historic condition of a site is necessary for establishing project goals. In discussions with community groups interested in restoration, their enthusiasm for tree-planting or prairie burns often leads to the development of action plans before three basic questions are asked: *What do we have? What is wrong with it? What can we do about it?*
2. Clearly articulate project goals that are appropriate to the ecology of your site. This is critical both for planning restoration action and selecting monitoring techniques. *One of the goals of the Cootes Paradise restoration is to improve the "quality" (area, density and diversity) of degraded plant communities*. This statement then guides implementation and monitoring of the restoration, and allows for meaningful evaluation. Earlier discussions about the restoration included "increasing emergent plant cover to 85% of the marsh area" as a goal. This is inappropriate in view of the natural fluctuations in wetland vegetation in response to changing water levels.
3. Restoration and monitoring programs need to be flexible enough to incorporate future refinement. Restoration ecology will never develop to the point where it can be said with certainty that action A will alter condition B so that site C will become a "pristine" natural space. Anticipate modifying your approach to address confounding effects D through Z! This is equally important for monitoring and implementing restoration programs.

References

Chow-Fraser, P., V. Lougheed, V. Le Thiec, B. Crosbie, L. Simser, and J. Lord. 1998. Long-term response of the biotic community to fluctuating water levels and changes in water quality in Cootes Paradise Marsh, a degraded coastal wetland of Lake Ontario. *Wetlands Ecology and Management* 6:19-42.

Chow-Fraser, P. and L. Lukasik. 1995. Cootes Paradise Marsh: Community participation in the restoration of a Great Lakes coastal wetland. *Restoration and Management Notes* 13(2):183-189.

Hahn, D. 1997. Passive Fishwaste Composting PCB Destruction Project. Ontario Ministry of Energy and Environment, Waste Reduction Branch, April 29, 1997. Memorandum on file at RBG.

Judd, W.W. 1950. Plants collected in the Dundas Marsh, Hamilton, Ont., 1946. *Canadian Field-Naturalist* 65:127-129.

Keddy, P.A. and A.A. Reznicek. 1986. Great Lakes vegetation dynamics: the role of fluctuating water levels and buried seeds. *Journal of Great Lakes Research* 12(1):25-36.

Lavender, B. 1987. Historical Geography of Cootes Paradise, Ontario. Senior Honours

Thesis, Department of Geography, Faculty of Environmental Resource Studies, University of Waterloo. Waterloo, Ontario, Canada.

Lougheed, V.L., B. Crosbie and P. Chow-Fraser. 1998. Predictions on the effect of common carp (Cyprinus carpio) exclusion on water quality, zooplankton, and submergent macrophytes in a Great Lakes wetland. *Canadian Journal of Fisheries and Aquatic Sciences* 55:1189-1197.

Lundholm, J.T. and W.L. Simser. 1999. Regeneration of submerged macrophyte populations in a disturbed Lake Ontario coastal marsh. *Journal of Great Lakes Research* 25 (2):395-400.

Painter, D.S., K.J. McCabe and W.L. Simser. 1989. *Past and present limnological conditions in Cootes Paradise affecting aquatic vegetation*. Royal Botanical Gardens Technical Bulletin No. 13. Hamilton, Ontario: Royal Botanical Gardens.

Pomfret, B. and J. Benkhuysen. 1993. Royal Botanical Gardens Ecology Department Progress Report 1993 Wildlife and Vegetation Monitoring Results. Unpublished report on file at Royal Botanical Gardens, Hamilton, Ontario.

Simser, W.L. 1982. *Changes in the aquatic biota of Cootes Paradise marsh*. Royal Botanical Gardens Technical Bulletin No. 12. Hamilton, Ontario: Royal Botanical Gardens.

Trotter, K., J.D. Hall and V. Cairns. 1998. Fish & wildlife habitat restoration project general information fact sheet. The fish & habitat restoration project. Hamilton, Ontario.

van der Valk, A.G. 1981. Succession in wetlands: A Gleasonian approach. *Ecology* 62(3): 688-696.

Van Horn, M. and K. Van Horn. 1996. Quantitative photomonitoring for restoration projects. *Restoration and Management Notes* 14(1):30-34.

Westcott, K., T.H. Whillans and M.G. Fox. 1997. Viability and abundance of seeds of submerged macrophytes in the sediment of disturbed and reference shoreline marshes in Lake Ontario. *Canadian Journal of Botany* 75:451-456.

Whillans, T.H. 1996. Historic and comparative perspectives on rehabilitation of marshes as habitat for fish in the lower Great Lakes basin. *Canadian Journal of Fisheries and Aquatic Sciences* 53(Suppl. 1):58-66.

Wragg, L.E. 1949. The Muskrat in Southern Ontario. M.A. Thesis, University of Toronto. 148 pp.

Tyler Smith is a Field Botanist at Royal Botanical Gardens, P.O. Box 399, Hamilton, ON, L8N 3H8, 905/527-1158 ext. 238, FAX 905/577-0375, tsmith@rbg.ca

Jeremy Lundholm is a Ph.D. candidate in the Botany Department, University of Guelph, Guelph, ON, N1G 2W1, 519/824-4120 ext. 6008, panax@sympatico.ca

Len Simser is the Natural Land Steward at Royal Botanical Gardens, P.O. Box 399, Hamilton, ON, L8N 3H8, 905/527-1158 ext. 238, FAX 905/577-0375, lsimser@rbg.ca

---